

JEE MAIN REVISION MASTER: ATOMIC STRUCTURE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Quantum Numbers & Electronic Configuration	6 PYQs	Quantum Numbers
Moderate	Heisenberg Uncertainty & De Broglie Wavelength	4 PYQs	Wave Mechanics
Standard	Bohr's Orbitals & Electronic Transitions	2 PYQs	Bohr Model

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Quantum Numbers & Electronic Configuration	$\text{Radial nodes} = n - l - 1$, $\text{Angular nodes} = l$, $\text{Total nodes} = n - 1$, $\mu_s = \sqrt{n(n+2)} \text{ BM}$
Heisenberg Uncertainty & De Broglie Wavelength	$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$, $\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$
Bohr's Orbitals & Electronic Transitions	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$, $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Quantum Numbers & Electronic Configuration (Moderate Yield — 6 PYQs)

Core Principle: Orbital shapes, quantum numbers (n, l, m_l, m_s), nodal surfaces, and anomalous configurations (Cr: $3d^5 4s^1$, Cu: $3d^{10} 4s^1$).

■ **Caution / JEE Pitfall:** Spin-only magnetic moment $\mu = \sqrt{n(n+2)}$ BM depends on n , which is the number of UNPAIRED electrons, not total electrons!

■ **Speed Trick:** Degeneracy of hydrogen atom at energy level n is n^2 (without spin) or $2n^2$ (including spin).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Which of the following is the CORRECT statement about Ψ^2 ?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of orbitals associated with 'N' shell of an atom is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The energy associated with first orbit of He^+ is..." [Ans: N/A]

3.2 Heisenberg Uncertainty & De Broglie Wavelength (Moderate Yield — 4 PYQs)

Core Principle: Limits of simultaneous precision in position and momentum, and dual wave nature of micro-particles.

■ **Caution / JEE Pitfall:** If uncertainty in position and momentum are equal ($\Delta x = \Delta p$), then $\Delta v = (1/2m)\sqrt{(h/\pi)}$.

■ **Speed Trick:** De Broglie wavelength of an electron accelerated through potential V volts: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Which of the following represents de Broglie equation?..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "According to the Heisenberg's Uncertainty principle, the value of $\Delta b \cdot \Delta x$ for an object whose mass is 10^{-6} kg is $(h=6.626 \times 10^{-34})$ " [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "If wavelength of photon is $2.2 \times 10^{-11} \text{ m}$ and $E=6.6 \times 10^{-34} \text{ Js}$, then momentum of photon..." [Ans: N/A]

3.3 Bohr's Orbitals & Electronic Transitions (Standard Yield — 2 PYQs)

Core Principle: Electronic energy levels in hydrogenic species, wave number $\bar{\nu}$, and emission spectral series.

■ **Caution / JEE Pitfall:** Rydberg formula requires multiplication by Z^2 for single-electron ions like He^+ ($Z=2$) and Li^{2+} ($Z=3$).

■ **Speed Trick:** Limiting line (series limit) of any spectral series occurs when $n_2 = \infty$ implies $\bar{\nu}_{\text{limit}} = (R Z^2 / n_1^2)$.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "The number of protons, neutrons and electrons in the ion $(^{16})^{32}\text{S}^{2-}$ respectively are..." [Ans: N/A]
- [2020 Shift] Q#1 (MCQ): "With regard to photoelectric effect, identify the correct statement among the following...." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Which of the following is the CORRECT statement about Ψ^2 ?

- (A) Ψ^2 represents atomic orbit
- (B) Probability density of the electron at that point
- (C) $\Psi^2 \neq 0$ for nodes
- (D) Ψ^2 has no physical meaning

Correct Answer: None

Solution Analysis: The correct statement is: Option B: Probability density of the electron at that point In quantum mechanics, Ψ is the wave function of an electron. The wave function itself, Ψ , does not have direct physical meaning. But the square of the wave function gives the probability density of finding the electron at a particular point. More correctly, we write it as: $|\Psi|^2$ For real wave functions, this is often written simply as: Ψ^2 So, Ψ^2 tells us the probability density of the electron at that point. Also, at nodes, the probability of finding the electron is zero, so:

Benchmark Problem 2: KCET 2026 (Q#2)

Which of the following represents de Broglie equation?

- (A) $\lambda = \frac{h}{\sqrt{mv}}$
- (B) $\lambda = (h / mv)$
- (C) $\lambda = (h / mp)$
- (D) $\lambda = (h / p)$

Correct Answer: None

Solution Analysis: The de Broglie equation is $\lambda = (h / p)$ where λ is the wavelength, h is Planck's constant, and p is the momentum of the particle. For a particle of mass m moving with velocity v , $p = mv$. So, $\lambda = (h / mv)$. Hence, the correct option is: Option B $\lambda = (h / mv)$

Benchmark Problem 3: KCET 2023 (Q#2)

The number of protons, neutrons and electrons in the ion ${}_{16}^{32}\text{S}^{2-}$ respectively are

- (A) 16,18,16
- (B) 16,16,18
- (C) 18,16,16
- (D) 16,16,16

Correct Answer: None

Solution Analysis: To find the number of protons, neutrons, and electrons in the sulfur ion ${}_{16}^{32}\text{S}^{2-}$, we can analyze the given atomic symbol: The atomic number (lower subscript) corresponds to the number of protons. In this notation, the atomic number is 16, so sulfur has 16 protons. The mass number (upper superscript) is the sum of protons and neutrons in the nucleus. Here, the mass number is 32. Since we already know there are 16 protons, we can find the number of neutrons by subtracting the number of protons from the mass number: $\text{Number of neutrons} = \text{Mass number} - \text{Number of protons}$. So, $\text{Number of neutrons} = 32 - 16 = 16$. Since the ion is S^{2-} , it has 2 extra electrons. Therefore, the number of electrons is $16 + 2 = 18$. The correct option is (A) 16,18,16.